

The Manager Over and Underperformance (MOU) Index: Ranking Soccer Managers

Soccer Track
20251420

1. Introduction

Data analytics in soccer has seen significant development in recent years. Most elite football clubs employ data analytics departments and many use data to motivate their decisions. The product that is delivered to fans has also changed, with pundits and commentators now regularly citing advanced statistics in broadcasts. In this time many statistics have been developed to judge player performance, but comparatively few to analyze team managers.

The team manager represents an important role in soccer.¹ While squads are composed of over twenty individual players, each team only employs one team manager. The team's decision making is often centralized around this individual, more so than in sports such as baseball or American football. The role of these managers is multifaceted, including not only squad decision making but being a representative of the club, handling media duties, instilling a culture within the organization, and the maintenance of the egos of both their playing and coaching staff. The most important job for a manager, however, is to lead a team to success by maximizing their performance.

This paper seeks to quantify the ability of managers to improve the performance of their players above and beyond expectation. To this end, I develop a novel index based on advanced soccer statistics to directly compare and assess a manager's capacity to induce player performance at both the squad level and the position group level. I generate a predictive algorithm to predict player performance and compare this with their actual performance to evaluate whether a manager is consistently able to generate higher performance from players. This index represents a novel contribution to the literature on managerial analysis in sports and is one of the first such models to integrate advanced analytics, direct manager comparisons, and the ability to evaluate managers by position groups.

Existing literature on the impact of managers on subordinate performance in wider labor economics is sparse. In general, research looking at managers has focused on high level organizational leaders such as CEOs (Bennedson, Perez-Gonzalez, & Wolfenzon, 2007; Bennedson, Perez-Gonzalez, & Wolfenzon, 2012) or political leadership (Jones and Olken, 2005) instead of middle management. Papers that have looked more directly at these mid-level bosses have found productivity boosts associated with switching from low quality to high quality managers (Lazear et al., 2015). Most research in this domain has been conducted in sports. In the literature on sports managers, studies have consistently found that there are differences between top rated and bottom rated managers on performance (Muehlheusser et al., 2018).

¹ Some organizations in soccer refer to the tactical leader of their team as a manager, while others refer to them as head coaches. This paper will default to referring to these individuals as managers, although these terms are often interchangeable.

Throughout the literature, there have been attempts to control for the impact of team quality on success to isolate manager effects. Early analysis focused on the development of a statistic for managerial efficiency and its ability to predict survivability in baseball, providing a basis for the literature (Porter and Scully, 1982; Scully, 1994). These earlier studies in the literature focused on the impact of management on winning percentages and extending their tenure (Kahn, 1993; Scully, 1994). Some studies approached this problem by using stochastic frontier production function analyses from Aigner et al. (1977) to predict optimal team performance (Bridgewater, Kahn, and Goodall, 2011; Dawson & Dobson, 2002; Frick and Simmons, 2008). This prediction is most often based on team financial records (Kahn, 1993; Buzzacchi et al., 2021; Frick and Simmons, 2008; Dawson, Dobson, and Gerrard, 2000). In these studies, it has been found that managers who were high level athletes themselves tend to perform well, as do managers who are highly experienced in management (Bridgewater, Kahn, and Goodall, 2011). Additionally, managers can achieve higher points totals by reducing inefficiencies (Frick and Simmons, 2008).

The research that has been conducted into managerial effects thus far has focused on the determinants of successful managers. On average, managers tend to have a significant influence on sporting and financial performance (Buzzacchi et al., 2021), with some estimating that 19% of team performance variation can be attributed to managers (Anderson and Sally, 2013). While this is lower than some may hope or expect, the existence of any managerial effect provides the opportunity for teams to optimize performance through appropriate manager selection. Managers with significant experience both as a player and a manager can increase the winning percentage and playoff success of their teams (Goodall, Kahn, and Oswald, 2011; Bridgewater, Kahn, and Goodall, 2011). This finding holds across sports, and has been found in basketball (Berri et al., 2009) and baseball (Kahn, 1993).

Managerial turnover represents a significant hazard for managers, with short term fluctuations in form often resulting in early termination in a field with already short average tenure, emphasizing the need for constant success (Audas, Dobson, and Goddard, 2009; Barros, Frick, and Passos, 2009). The impact of this termination in improving performance, however, has been found to be small (Bryson et al., 2024). As such, there is constant pressure on managers to improve performance.

Many of these studies, however, have flaws in their assessment of manager performance. While they correctly try to control team quality, their outcome variables are imperfect measures of team performance. Assessing a manager based on points or winning percentage is logical in sports with a high sample size of games or high scores such as basketball and baseball as teams will have a higher chance of achieving their true value over the course of a season. In a low scoring, low sample size game like soccer, on the other hand, this same analysis is prone to error. Looking, for example, at the regular discrepancy between the true league table and the expected points table (xPTS) indicates that while performance is often correlated with winning, the correlation is imperfect.² The use of these variables in past studies is due to the lack of data availability, but with the advances in sports statistics we are now able to more accurately assess team performance.

² The xPTS table or “Justice” table is a tool to analyze team performance. It calculates the probability of a team winning, drawing, or losing a game based on the expected goals (xG) they generate in a match compared to their opponent, and multiplies these probabilities by the points totals (3 for a win, 1 for a draw, 0 for a loss) to find the amount of points a team has “earned” in a match. These points earned in a season can be divergent from true points totals. Justice tables for the Top 5 Leagues considered in this paper can be found on [Understat.com](https://understat.com).

Additionally, existing research is focused on aggregates. This is true in two senses, both looking at aggregates at the managerial level (i.e top vs bottom managers) and looking at teams at the aggregate instead of assessing them as groups of individual units (i.e defenders vs midfielders). This focus on group analysis simplifies the analysis to answer the question of managerial effectiveness but loses granularity through which we can compare managers. Some managers may be specialists instead of generalists, excellent at improving certain players but not others. Additionally, it is valuable to explore the full distribution of managers as done in Berri et al., (2009) to better understand relative performance and to challenge conventional beliefs surrounding the quality of certain figures.

To answer these gaps in the literature and to advance the statistics regarding the assessment of coaching, this paper introduces the Manager Over and Underperformance (MOU) Index.³ Following from the work of Berri et al., (2009), the MOU index looks at the deviation of a set of players from their expected performance under a manager instead of their impact on wins or points. Looking at these deviations on an individual level allows for easier analysis of different subsections of teams, giving a more holistic view of a manager's qualities and strengths. This index also generates a single statistic through which managers can be directly compared, which is calibrated against manager Elo. This index is the first of its kind to leverage the newly developed wealth of soccer statistics. It provides a general formula for easy replication and development, as well as a potential outcome variable through which future studies may be able to better understand the impact of managerial qualities on performance.

The paper is organized as follows: Section 2 introduces the methodology used in generating the MOU index, as well as an introduction to the data sources. Section 3 presents the main results of the paper by looking at the top and bottom performers on the index as well as attuning the index against manager Elo scores. Section 4 discusses the findings of the study, the strengths and limitations of the index, and potential avenues for future research using this methodology.

2. Methods

2.1. Estimation

While a soccer manager may have many responsibilities, the primary function of the manager is to lead a team to on-field success. While this can be defined as winning trophies or matches, teams must often form more realistic performance expectations based on their means and status. As a result, on-field success can be more broadly defined to claim that a successful manager is one who induces a team to consistently perform above their expectation. To accomplish this, a manager must make individual players, or groups of players, perform above their expected performance. It is from this assumption that I derive my central model in equation (1).

$$\delta_{mg} = \kappa \frac{x_g - \hat{x}_g}{\hat{x}_g} \quad (1)$$

³ This name is a backronym of "Mou," a popular nickname for Portuguese manager José Mourinho.

Where δ represents an index value for some manager m over a position group g . This δ is made up of two parts; firstly, the percentage difference in some statistic x from its predicted value $\hat{x}$, and secondly a dampening factor κ defined as

$$\kappa = \frac{1}{N - n + 1} \quad (2)$$

Where n is the number of season-level observations for a manager, and N is the total number of seasons in the sample. This value can then be interpreted as an “unemployment penalty,” reducing the overall value of a manager if they perform well in only one or two seasons, helping us to avoid anomalous observations.

The value x can be defined broadly to allow for replication with alternative statistics. The only requirements on x to form this index is that it represents some minutes weighted average of player performance. For the current analysis, it is defined as

$$x_g = \frac{\sum_{p,g,t} x_{pgt} \mu_{pgt}}{\sum_{p,g,t} \mu_{pgt}} \quad (3)$$

$$\hat{x}_g = \frac{\sum_{p,g,t} \hat{x}_{pgt} \mu_{pgt}}{\sum_{p,g,t} \mu_{pgt}} \quad (4)$$

Where the value x_g represents the sum of the individual level statistics x for all players p in position group g for season t , weighted by individual minutes played μ . Equally, the predicted value to which it is compared, $\hat{x}_g$ looks at a weighted average of individual level predictions. The generation of strong predictions is necessary to the development of a sufficient statistic, but this is up to the discretion of the analyst. The threshold for this prediction may change depending on the availability of data.

The major assumption of this model is that when controlling for player quality, the deviation in performance is due to managerial differences in the aggregate. While other factors may influence individual performance such as club dynamics, network effects with groups of players, and various extraneous factors influencing individual players, the variation in position group aggregate performance is primarily attributed to the direct influence of managers or indirect managerial effects.

2.2. Data and Testing

I tested this model in the context of men’s football in the “Top 5” European leagues from the beginning of the 2017/2018 season to the conclusion of the 2023/2024 season. Testing for this model was completed using publicly available data taken from various data sources. For the x in my model, I used the DAVIES statistic developed by American Soccer Analysis (ASA).⁴ DAVIES attempts to quantify a soccer player’s contribution to their team relative to a “standard” alternative, much like Wins Above Replacement in baseball or Player Efficiency in basketball. The developers of DAVIES first used a group of seventeen statistics to calculate a model for the expected Goals Added by a player

⁴ DAVIES is a backronym, standing for Determining Added Value of Individual Effectiveness including Style. It was named after Canadian footballer Alphonso Davies.

through their on-pitch contributions (i.e passing, shooting, dribbling, defending). It then clusters players based on their attributes and statistics into both broad and specific categories for playing roles. This allows for a more granular analysis of a player's contribution as they fulfil a specific niche, differentiating between individuals who may play the same position in different styles. These groups are then stratified by age and compared, creating a final statistic that reports the Goals Added by a player relative to a similar, average player who plays in the same role and has the same age.⁵

DAVIES was selected as the statistic of interest for this model due to its ability to directly compare across position groups and its advanced statistical backing. To date, it represents the most complete statistic to compare players across specializations using open source, frequently updated data. DAVIES data is reported at the season level for each player who played in the Top 5 leagues over the course of our dataset. Further DAVIES data is available for some alternative leagues and in top women's leagues, but the Top 5 European men's leagues were selected for this first analysis. This was due to the availability of statistics such as Manager Elo with which to benchmark the value, and the ability to generate sufficiently strong predicted values.

Supplemental data on player statistics and age are obtained from the website FBRef.com, the field's standard source for open-source data. This data is provided by Opta. FBRef.com also has estimated wage data provided by Capology. For the Top 5 leagues many of these wage statistics are confirmed, but some have been estimated. Estimations are made primarily for youth players or newly promoted teams. I interpolated wage statistics for players who had missing wages in order to create player predictions for all relevant players. More information on this interpolation can be found in the appendix A.2.

Manager tenure data was taken from Transfermarkt.com, which has appointment and dismissal dates for all teams in the dataset. A manager was attributed a team season if they took control of the team for greater than 50% of the days in the season (a season going from August 1st of one year to July 31st of the next). Days were selected above matches in charge to quantify manager tenure to consider the work managers are able to implement over periods with no matches such as international breaks and preseason training.⁶

The data is calibrated primarily by comparing it to an existing measure of manager quality, manager ELO taken from clubelo.com. This rating system uses a traditional ELO rating system like those used in Chess and other events.⁷ It awards points to individuals who win in direct competition with others, and subtracts from those who lose.⁸ The points awarded or subtracted from the original score is

⁵ A clear drawback of this statistic is its emphasis on goals added which will naturally favor attacking players in each positional group. A similar goals subtracted model is currently in development in ASA that can eventually be incorporated into the model.

⁶ Season matches normally run between mid-to-late August each year and mid-to-late May each year, depending on the league. With international tournaments taking place from mid-June to mid-July, and players taking time off, players will generally recommence training towards the end of July for the new season, leading to August 1st as a reasonable date for the season threshold.

⁷ ELO ratings are named after their creator Arpad Elo.

⁸ The algorithm used for manager ELO rankings is $\Delta ELO = (R - E) * K$ where R is the result (1 for win, 0.5 for draw, 0 for loss), K is a constant scalar in this case set to 20, and E is defined as the ex-

proportional to the relative status of the two competitors before the contest. The strength of ELO systems is that with sufficient sample size an ELO rating should approach a true value of quality. It is currently used by FIFA as the primary ranking system for international football teams. As a manager rating system it performs fairly well, but ELO ratings between managers are endogenous to team quality and stature which may artificially inflate or deflate manager ratings based on factors outside their control.

3. Results

The results are presented in three parts, firstly presenting the index, then a discussion of the prediction algorithm created for $\hat{x}$ using DAVIES, and finally comparing the model to the existing measure of manager ELO.

3.1. Manager Over and Underperformance Index

Overall, a MOU index value was generated for 241 managers representing 147 teams. The average number of attributed seasons to managers in the sample was 2.83. The MOU Index had a minimum value of -0.190, and a maximum of 0.350. The top and bottom 10 performing managers can be found in Table 1. A full ranking of the 241 managers in the sample can be found in appendix A.1.

Table 1: Ranking of the Top 10 and Bottom 10 managers according to MOU Index

	Manager	Rating		Manager	Rating
1	Jurgen Klopp	0.350	232	Frank Kramer	-0.050
2	Pep Guardiola	0.321	233	Steve Bruce	-0.050
3	Gian Piero Gasperini	0.286	234	Jan Siewert	-0.051
4	Simone Inzaghi	0.190	235	Stefan Leidl	-0.055
5	Stefano Pioli	0.165	236	Sergio	-0.057
6	Julian Nagelsmann	0.163	237	Olivier Pantaloni	-0.058
7	Roberto De Zerbi	0.084	238	Christophe Pelissier	-0.067
8	Bruno Genesio	0.069	239	Pepe Bordalas	-0.069
9	Adi Hutter	0.068	240	Christian Streich	-0.129
10	Juan Carlos Unzué	0.059	241	Sean Dyche	-0.190

Upon general inspection, the MOU index performs well at separating successful and unsuccessful managers. The top two managers on the index, Jurgen Klopp and Pep Guardiola, are widely considered to be elite managers in the modern game. Similarly, managers such as Julian Nagelsmann, Gian Piero Gasperini, and Roberto De Zerbi have received many plaudits for their ability in recent years. The performance of managers in the bottom 10 is not as clear. While none of these managers are considered top of the field, managers such as Sean Dyche and Pepe Bordalas have been modestly

ante win probability for a team defined as $E = \frac{1}{\frac{-dr}{(10^{400} + 1)}}$ with dr being the difference in ELO

between the two competitors before the contest.

successful in recent years for their clubs. The appearance of Sean Dyche and Steve Bruce on the section of the list points towards the list favoring attacking managers instead of those who set up defensively, which is a deficiency of the DAVIES statistic.⁹

A distribution of the MOU Index values can be found in Figure 1. The mean value of the MOU Index is 0.001 which is not significantly different from 0 ($t = 0.313$, $p = 0.754$, $CI = [-0.005, 0.007]$). The distribution is right skewed ($\tilde{\mu}_3 = 3.401$) and leptokurtic ($Kurt = 22.883$). This signifies that the distribution is marked by the existence of several outliers, particularly large positive outliers. This indicates that while many managers have little to no correlation with player performance, a number of managers are associated with either largely positive or largely negative deviations in those performances. This is in keeping with findings from other papers on the differential impact of top-level managers but the overall substitutability of managers in the field.

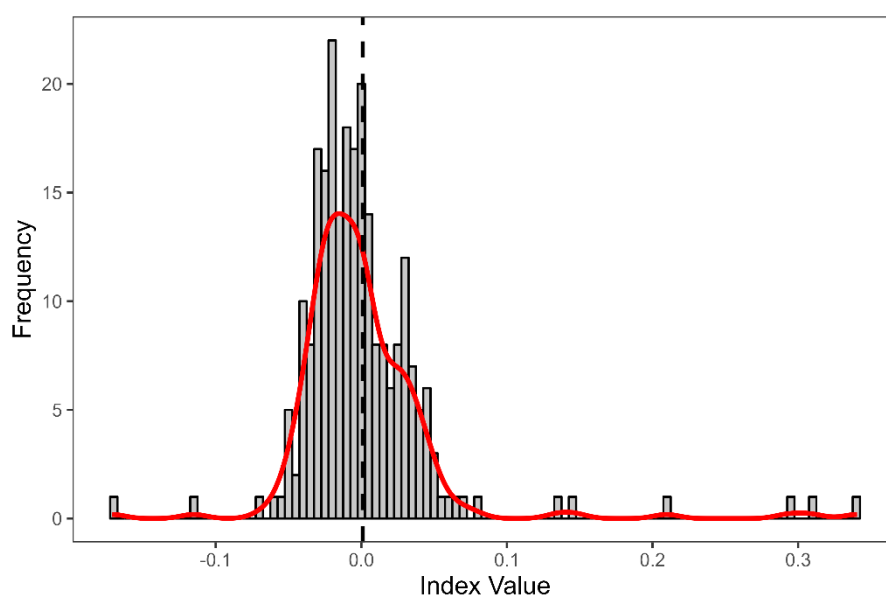


Figure 1: Distribution of MOU Index Values with Gaussian density function

The interpretation of MOU index values is not entirely clear due to the presence of the dampening factor κ . In the absence of this factor the MOU index would represent the percentage deviation in DAVIES attributed to each manager. However, with this factor involved the index should simply be interpreted as a standalone statistic.

The MOU Index also gives the ability to look at manager performances with certain position groups in order to both directly compare managers as well as to assess managerial strengths. Figure 2 compares the top 2 managers in the overall index, Jurgen Klopp and Pep Guardiola, on their relative strengths. The top and bottom 10 performing manager position groups can be found in Appendix A.3.¹⁰

⁹ Evidence for this can be found in Appendix B.1, where Sean Dyche appears to improve his defensive players, while having underperformances in other areas of the pitch.

¹⁰ The index shows great flexibility for analysis. Only a subset of possible permutations and applications are shown here as a proof of concept.

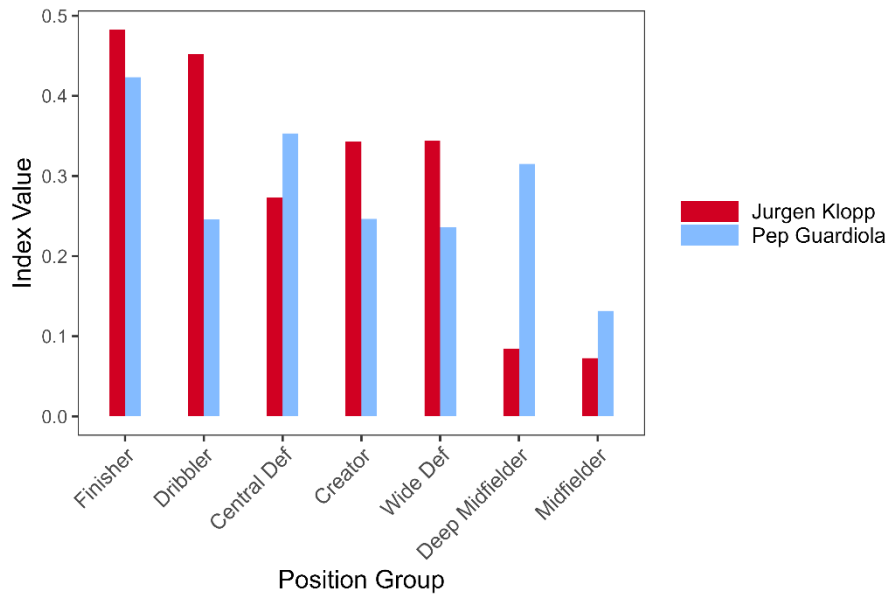


Figure 2: A position wise comparison of Jurgen Klopp and Pep Guardiola

3.2. Predicting Player Performance

The performance and success of the MOU Index is dependent on the choice and quality of the underlying prediction algorithm. The prediction algorithm chosen for the index presented in section 3.1 was the result of an iterative process. Four of the models considered for the index can be found in Table 2.

Table 2: Selected DAVIES Prediction Algorithms

	Dependent Variable:			
	(1)	(2)	(3)	(4)
WeeklyWageEUR	0.00001*** (0.00000)	0.00001*** (0.00000)	0.00001*** (0.00000)	0.00000*** (0.00000)
Age	-0.043*** (0.004)	-0.060*** (0.002)	-0.048*** (0.003)	-0.041*** (0.003)
Minutes		0.002*** (0.00001)	0.002*** (0.00001)	0.002*** (0.00002)
$DAVIES_{t-1}$			0.189*** (0.006)	0.162*** (0.008)
$DAVIES_{t-2}$				0.068*** (0.007)
League	Yes	Yes	Yes	Yes
Role	No	Yes	Yes	Yes
Constant	1.005*** (0.111)	-1.967*** (0.069)	-2.018*** (0.084)	-2.210*** (0.106)
Observations	14,196	13,811	9,756	6,784
R^2	0.132	0.728	0.762	0.773
Adjusted R^2	0.132	0.728	0.762	0.773
Resid. Std. Error	1.978 (df=14189)	1.108 (df=13797)	1.061 (df=9741)	1.050 (df = 6768)
F Statistic	359.934*** (df=6; 14189)	2,846.363*** (df=13; 13797)	2,232.285*** (df=14; 9741)	1,538.130*** (df=15; 6768)
<i>Notes: Leagues included are Ligue 1, La Liga, Premier League, Bundesliga, and Serie A. Roles included are Creator, Deep Midfielder, Dribbler, Finisher, Midfielder, Wide Defender, and Central Defender. These variables are included as dummies, not fixed effects specifications.</i>				

*p<0.1; **p<0.05; ***p<0.01

Regression 2 was selected as the eventual prediction algorithm for the final index. It was selected due to its combination of high R^2 value (Adjusted $R^2 = 0.728$) and number of observations (13,811). As can be seen in columns 1 and 3, the exclusion of variables included in model 2 significantly reduced

the R^2 of the model, and the inclusion of lagged statistics as in models 3 and 4 increased the R^2 of the model but at the cost of thousands of observations.¹¹

This choice of specification resulted in a pure prediction for 13,811 player seasons. The 385 observations lost between model 1 and model 2 were mainly players who had no logged minutes in a given season, which has no impact on analysis. There were 420 observations for whom predictions were not generated. This was mainly due to missing data on weekly wage. To not lose these players, their wage data was interpolated based on a prediction that is detailed in Appendix A.2. Figure 3 shows the relationship between predicted DAVIES and observed DAVIES. It is clear from this figure that the linear regression model is a strong fit for the observed data.

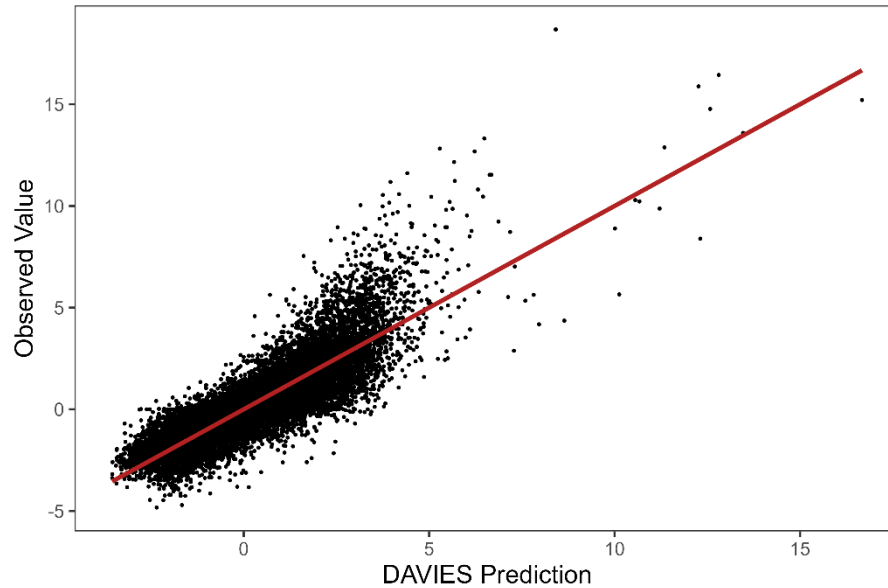


Figure 3: *DAVIES Prediction vs Observed Value with Line of Best Fit*

3.3. Comparison with manager ELO

To calibrate the MOU index beyond general inspection, it must be shown to perform well against existing measures of manager performance. As mentioned in Section 1 this data on individual manager performance is sparse, but some metrics do exist. I calibrate the model by comparing it to Manager Elo. The plot of the MOU Index against corresponding Manager Elo can be found in Figure 4. The Pearson correlation coefficient between these two variables was 0.4591, demonstrating a modest positive relationship between the two variables.

¹¹ The major driver of the prediction is the minutes variable. This makes sense, as the more minutes played gives more opportunities to add goals. Since the minutes played by players is a choice made by managers (except for in the instance of injuries), this model was preferred to an alternative specification with DAVIES per 90 minutes.

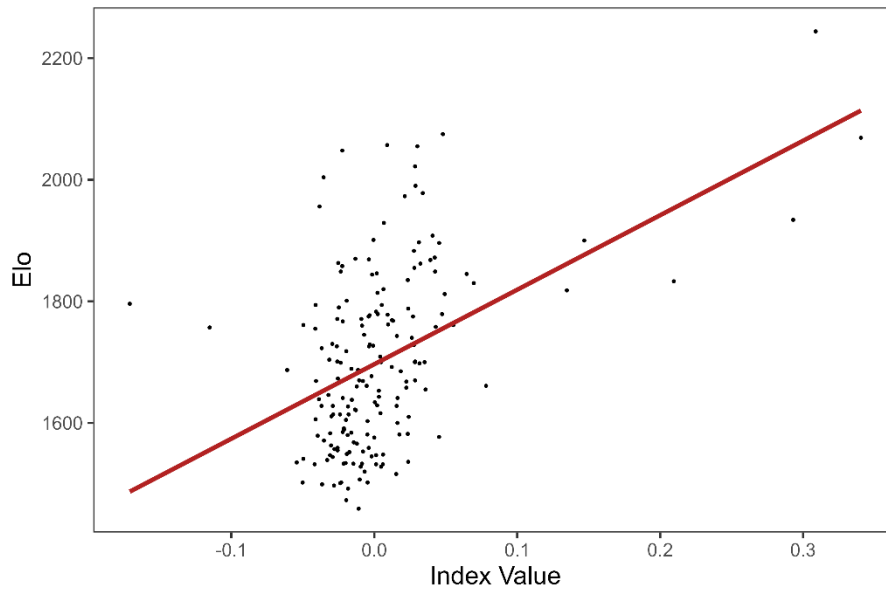


Figure 4: MOU Index value vs Manager Elo with Line of Best Fit

This positive correlation is to be expected and a clear sign of the functioning of these two models. Managers who tend to be higher on the index will also tend to have higher Elo ratings. An issue with Elo ratings, however, is that when applied to soccer they are prone to bias. In a game where Elo is commonly used such as chess, each player is given equal material. When comparing managers in soccer teams, however, teams can be quite unevenly matched for player quality. This means that comparatively weak (strong) managers can have biased Elo ratings if they manage comparatively strong (weak) teams.

This is exhibited in the variation of Elo that can be observed surrounding an index value of 0 in Figure 4. In a perfectly efficient labor market, the best managers would be sorted into the best positions based on merit and we would see both greater discrepancy in Elo ratings but also a higher correlation with the MOU index. The absence of this indicates inefficiency in the sorting of managers to positions.

While it would be prudent to calibrate the MOU index further, it is at this time not possible. As soccer data develops, there will be more ways to verify and compare this index to alternative measures of manager performance. Additionally, in time, the index will develop with the introduction of new data, and it will be possible to observe whether managers that score higher on the index tend to have more successful careers in management. This index represents a first attempt at the development of a usable statistic on individual managers, which will develop as the field progresses.

4. Discussion

The development of the Manager Over and Underperformance Index represents the first of its kind in soccer literature, and one of the only existing statistics to quantify managerial performance. As a metric, the MOU index indicates that there is high substitutability between most top-level managers, in keeping with other findings in the literature.. It is also correlated quite highly with existing metrics of managerial performance such as Manager ELO, while offering many advantages over these alternative statistics. This paper represents a novel contribution to soccer statistics and provides a

new way to analyze managerial effects, predict a manager's success, and rate their overall performance.

The MOU Index has several attributes that improve upon existing statistics. Firstly, it attempts to address the issue of endogeneity in existing metrics by controlling for player quality. Other forms of judging managers, such as points per game, manager ELO, or team success, are intrinsically linked to the players at a manager's disposal but unrelated to the manager's quality themselves. The algorithm underlying the MOU Index addresses this by measuring deviation from predicted player performance, allowing for direct comparison for managers between clubs.

The MOU Index, while here constructed with a specific statistic, represents instead a broad algorithm for the judgement of managers. The formula is generalizable to other statistics or data sources. It is also malleable to advances in player statistics, allowing it to evolve over time as the judgement of player performance improves.

The data requirements for the MOU Index ensure its transparency and replicability. It uses exclusively easy access open-source data. This data is also provided by several of the most trusted data providers in the sport in Opta and Transfermarkt. Due to the open-source nature of this dataset and the constantly developing state of statistics in world football, the MOU index can expect to only improve in future iterations as the field develops.

Additionally, the index performs well in predicting managerial performance based on its correlation with ELO and general inspection. Observing whether the managers with higher indices rise to the top of the field in the coming years will further test the quality of its predictions.

As a new development in the field, the index does come with drawbacks that may be improved in time. Firstly, the model makes several assumptions regarding the operation of clubs. It assumes that any excess deviations in player performance from expected is due to managerial effects. In truth, player deviation from expectation could be a result of a number of extraneous factors: media influence, returns from injury, cultural differences with players moving to different countries, internal politics, as well as other immeasurable factors. I contend that these are all potential factors influencing player performance that can be mediated by more effective management, and as such we can still attribute this deviation for the most part to managerial quality.

Another critique of the model is that managers may be effective through alternative channels. A manager may have value to a club above and beyond their impact on player performance down to how they interact with the media, how they develop and represent the culture of a club, the commercial draw that a big name or well-liked manager may bring to an organization, or the network effects that may come with former players wanting to work with the manager again and moving to a club that they would have not otherwise. This critique is valid and appears to matter anecdotally in world football, however these effects are largely unquantifiable. If statistics exist to quantify and compare managers on these aspects, the general form of the MOU Index allows for them to be easily integrated to the model. While the index presented in this current paper does not consider these factors, it does not necessarily exclude them from managerial analysis in the future.

The MOU Index represents a new paradigm in the world of soccer analytics, and a new way of analyzing aspects of the game. While it can be used to compare and contrast managers and assess performance, it also opens up the opportunity for future research. The MOU index could be used in

the future to try and determine which manager qualities best predict ability such as playing experience, age, or more advanced techniques like natural language processing of weekly press conferences. It has the capacity to look at the relative strengths of managers through their index for different position groups, thus increasing the information that employers may have and reducing current inefficiencies in labor markets. Similarly, it may allow clubs to better match their manager choices to the qualities of their squad to improve performance without the need for significant further investment.

Overall, the Manager Under and Overperformance Index represents a new way of analyzing football. As statistics improve in the sport, teams and commentators will seek more clarity from data to quantify team, player, and manager performance in a single statistic. While significant strides have been made in this direction on the player and team side in recent years, the MOU index achieves the same goal for managers.

References

- Aigner, D., Lovell, C. K., & Schmidt, P. (1977). Formulation and estimation of stochastic frontier production function models. *Journal of econometrics*, 6(1), 21-37.
- Anderson, C., & Sally, D. (2013). *The numbers game: Why everything you know about soccer is wrong*. Penguin.
- Audas, R., Dobson, S., & Goddard, J. (1999). Organizational performance and managerial turnover. *Managerial and Decision Economics*, 20(6), 305-318.
- Barros, C. P., Frick, B., & Passos, J. (2009). Coaching for survival: The hazards of head coach careers in the German 'Bundesliga'. *Applied Economics*, 41(25), 3303-3311.
- Bennedsen, M., Perez-Gonzalez, F., & Wolfenzon, D. (2007). *Do CEOs Matter?* (No. 13-2007). Working paper.
- Bennedsen, M., Pérez-González, F., & Wolfenzon, D. (2012). Estimating the value of the boss: Evidence from CEO hospitalization events. *Unpublished working paper. Columbia Business School, New York, NY*.
- Berri, D. J., Leeds, M., Leeds, E. M., & Mondello, M. (2009). The role of managers in team performance. *International Journal of Sport Finance*, 4(2), 75-93.
- Bridgewater, S., Kahn, L. M., & Goodall, A. H. (2011). Substitution and complementarity between managers and subordinates: Evidence from British football. *Labour Economics*, 18(3), 275-286.
- Bryson, A., Buraimo, B., Farnell, A., & Simmons, R. (2024). Special ones? The effect of head coaches on football team performance. *Scottish Journal of Political Economy*, 71(3), 295-322.
- Buzzacchi, L., Caviggioli, F., Milone, F. L., & Scotti, D. (2021). Impact and efficiency ranking of football managers in the Italian Serie A: Sport and financial performance. *Journal of Sports Economics*, 22(7), 744-776.

- Dawson, P., & Dobson, S. (2002). Managerial efficiency and human capital: an application to English association football. *Managerial and Decision Economics*, 23(8), 471-486.
- Dawson, P., Dobson, S., & Gerrard, B. (2000). Estimating coaching efficiency in professional team sports: Evidence from English association football. *Scottish Journal of Political Economy*, 47(4), 399-421.
- Frick, B., & Simmons, R. (2008). The impact of managerial quality on organizational performance: Evidence from German soccer. *Managerial and Decision Economics*, 29(7), 593-600.
- Goodall, A. H., Kahn, L. M., & Oswald, A. J. (2011). Why do leaders matter? A study of expert knowledge in a superstar setting. *Journal of Economic Behavior & Organization*, 77(3), 265-284.
- Jones, B. F., & Olken, B. A. (2005). Do leaders matter? National leadership and growth since World War II. *The Quarterly Journal of Economics*, 120(3), 835-864.
- Kahn, L. M. (1993). Managerial quality, team success, and individual player performance in major league baseball. *ILR Review*, 46(3), 531-547.
- Lazear, E. P., Shaw, K. L., & Stanton, C. T. (2015). The value of bosses. *Journal of Labor Economics*, 33(4), 823-861.
- Muehlheusser, G., Schneemann, S., Sliwka, D., & Wallmeier, N. (2018). The contribution of managers to organizational success: Evidence from German soccer. *Journal of Sports Economics*, 19(6), 786-819.
- Porter, P. K., & Scully, G. W. (1982). Measuring managerial efficiency: The case of baseball. *Southern Economic Journal*, 642-650.
- Scully, G. W. (1994). Managerial efficiency and survivability in professional team sports. *Managerial and Decision Economics*, 15(5), 403-411.

Appendix

Appendix A.1: Ranking of All Managers by MOU Index

Manager	Rating	Manager	Rating	Manager	Rating
1 Jurgen Klopp	0.350	31 Manuel Pellegrini	0.034	61 Giuseppe Iachini	0.016
2 Pep Guardiola	0.321	32 Robert Moreno	0.033	62 Marcelo Bielsa	0.016
3 Gian Piero Gasperini	0.286	33 Mauricio Pochettino	0.032	63 Paco Lopez	0.016
4 Simone Inzaghi	0.190	34 Michel Der Zakarian	0.032	64 Alfred Schreuder	0.015
5 Stefano Pioli	0.165	35 Marcelino	0.031	65 Frederic Hantz	0.014
6 Julian Nagelsmann	0.163	36 Luciano Spalletti	0.030	66 Oliver Glasner	0.013
7 Roberto De Zerbi	0.084	37 Carlo Ancelotti	0.029	67 Niko Kovac	0.012
8 Bruno Genesio	0.069	38 Pablo Machin	0.029	68 Christian Gourcuff	0.012
9 Adi Hutter	0.068	39 Andre Breitenreiter	0.029	69 Julen Lopetegui	0.012
10 Juan Carlos Unzué	0.059	40 Zinedine Zidane	0.029	70 Dieter Hecking	0.010
11 Paolo Fonseca	0.055	41 Pellegrino Matarazzo	0.028	71 Thomas Frank	0.009
12 Hansi Flick	0.050	42 Eddie Howe	0.028	72 Juan Muniz	0.009
13 Sebastian Hoeness	0.049	43 Ernesto Valverde	0.028	73 Unai Emery	0.009
14 Antonio Conte	0.048	44 Gerardo Seoane	0.028	74 Vincent Hognon	0.008
15 Franck Haise	0.047	45 Rudi Garcia	0.027	75 Vicente Moreno	0.007
16 Thomas Tuchel	0.045	46 Andoni Iraola	0.026	76 Xavi	0.007
17 Peter Bosz	0.045	47 Stephane Jobard	0.026	77 Jose Luis Mendilibar	0.006
18 Jupp Heynckes	0.045	48 Igor Tudor	0.024	78 Achim Beierlorzer	0.006
19 Ange Postecoglou	0.043	49 Philippe Montanier	0.024	79 Olivier Dall'Oglio	0.006
20 Lucien Favre	0.043	50 Gennaro Gattuso	0.024	80 Sinisa Mihajlovic	0.005
21 Laurent Blanc	0.042	51 Maurizio Sarri	0.023	81 Jesse Marsch	0.005
22 Edin Terzic	0.041	52 Eusebio Di Francesco	0.023	82 Alessio Dionisi	0.005
23 Marco Rose	0.039	53 Michel	0.022	83 Eusebio	0.005
24 Will Still	0.039	54 Philippe Clement	0.022	84 Quique Setien	0.004
25 Mikel Arteta	0.038	55 Gabriele Cioffi	0.022	85 Raffaele Palladino	0.004
26 Alessio Lisci	0.037	56 Ralf Rangnick	0.021	86 Rubi	0.004
27 Arsene Wenger	0.037	57 Peter Stoger	0.019	87 Patrick Vieira	0.003
28 Julien Stephan	0.036	58 Andrea Sottit	0.018	88 Domenico Tedesco	0.003
29 Xabi Alonso	0.035	59 Cuco Ziganda	0.017	89 Erik ten Hag	0.002
30 Pierre Sage	0.035	60 Frank Lampard	0.016	90 Andre Villas-Boas	0.002

91	Steffen Baumgart	0.002	126	Patrice Garande	-0.008	161	Asier Garitano	-0.019
92	Nuno Espirito Santo	0.002	127	Vahid Halilhodzic	-0.008	162	Albert Celades	-0.019
93	Carles Martinez	0.001	128	Ole Werner	-0.008	163	David Wagner	-0.019
94	Alain Casanova	0.001	129	Antoine Kombouare	-0.009	164	Walter Mazzarri	-0.019
95	Brendan Rodgers	0.001	130	Roberto D'Aversa	-0.009	165	Roberto Donadoni	-0.019
96	Claude Puel	0.001	131	Rob Edwards	-0.009	166	Thiago Motta	-0.020
97	Gustavo Poyet	0.001	132	Laszlo Boloni	-0.009	167	Luka Elsner	-0.020
98	Graham Potter	0.000	133	Jean-Louis Gasset	-0.010	168	Tayfun Korkut	-0.020
99	Andrea Pirlo	0.000	134	Daniele De Rossi	-0.010	169	Sabri Lamouchi	-0.020
100	Xavier Pimienta	0.000	135	Marco Giampaolo	-0.010	170	Pacheta	-0.021
101	Dino Toppmoller	0.000	136	Jagoba Arrasate	-0.011	171	Leonardo Semplici	-0.021
102	Luis Enrique	-0.001	137	Gary O'Neil	-0.011	172	Ruben Baraja	-0.021
103	Eric Roy	-0.001	138	Diego Lopez	-0.011	173	Walter Zenga	-0.022
104	Imanol Alguacil	-0.002	139	Leonardo Jardim	-0.011	174	Dean Smith	-0.022
105	Francesco Farioli	-0.002	140	Abelardo	-0.011	175	Quique Flores	-0.022
106	Luis Garcia	-0.002	141	Ole Gunnar Solskjaer	-0.012	176	Pal Dardai	-0.022
107	Francisco Rodriguez	-0.002	142	Stefan Ruthenbeck	-0.012	177	Patrick Kisnorbo	-0.022
108	Bo Svensson	-0.003	143	Jess Thorup	-0.012	178	Jose Mourinho	-0.022
109	Cesare Prandelli	-0.003	144	Uwe Rosler	-0.013	179	Ralph Hasenhuttl	-0.022
110	Jorge Sampaoli	-0.003	145	Marco Silva	-0.013	180	Pascal Dupraz	-0.023
111	Ronald Koeman	-0.004	146	Stephane Moulin	-0.014	181	Sam Allardyce	-0.023
112	Christophe Galtier	-0.004	147	Jocelyn Gourvennec	-0.014	182	Neil Warnock	-0.024
113	Heiko Herrlich	-0.004	148	Giovanni Stroppa	-0.015	183	Luca Gotti	-0.024
114	Steven Gerrard	-0.004	149	Javi Gracia	-0.015	184	Thierry Laurey	-0.024
115	Bernard Blaquart	-0.004	150	Bruno Labbadia	-0.016	185	Davide Nicola	-0.024
116	Vladimir Petkovic	-0.005	151	Manuel Baum	-0.016	186	Diego Martinez	-0.025
117	Daniel Farke	-0.005	152	Frank Schmidt	-0.016	187	Javier Aguirre	-0.025
118	Paulo Sousa	-0.005	153	Vincenzo Italiano	-0.016	188	Davide Ballardini	-0.026
119	Markus Gisdol	-0.005	154	Aurelio Andreazzoli	-0.016	189	Rolando Maran	-0.026
120	Fabio Liverani	-0.007	155	Regis Le Bris	-0.017	190	Florian Kohfeldt	-0.026
121	Eduardo Coudet	-0.007	156	Gerald Baticle	-0.017	191	Massimiliano Allegri	-0.026
122	Rafael Benitez	-0.007	157	Carlos Carvalhal	-0.017	192	Mauricio Pellegrino	-0.026
123	Bruno Lage	-0.007	158	Scott Parker	-0.018	193	Ivan Juric	-0.026
124	David Arteaga	-0.007	159	Thomas Letsch	-0.018	194	Oscar Garcia	-0.027
125	Fabio Liverani	-0.007	160	Friedhelm Funkel	-0.018	195	Marco Zaffaroni	-0.027

196	David Guion	-0.028	212	Gaizka Garitano	-0.032	228	Urs Fischer	-0.041
197	Jean-Louis Garcia	-0.028	213	Fabio Pecchia	-0.033	229	Enrico Maassen	-0.042
198	Martin Schmidt	-0.028	214	David Linares	-0.035	230	Markus Weinzierl	-0.042
199	Marco Baroni	-0.029	215	Sebastien Desmazeau	-0.035	231	Fabien Mercadal	-0.046
200	Nigel Pearson	-0.029	216	Uwe Neuhaus	-0.035	232	Michael Kollner	-0.046
201	Paolo Zanetti	-0.029	217	Diego Simeone	-0.035	233	Sergio	-0.048
202	Roy Hodgson	-0.029	218	Torsten Lieberknecht	-0.037	234	Steve Bruce	-0.050
203	Vincent Kompany	-0.030	219	Steve Cooper	-0.037	235	Olivier Pantaloni	-0.050
204	Domenico Di Carlo	-0.030	220	Thomas Reis	-0.037	236	Frank Kramer	-0.050
205	Pascal Gastien	-0.030	221	Claudio Ranieri	-0.038	237	Jan Siewert	-0.051
206	Alberto Gilardino	-0.030	222	Chris Hughton	-0.039	238	Stefan Leitl	-0.054
207	Sandro Schwarz	-0.031	223	Alvaro Cervera	-0.040	239	Christophe Pelissier	-0.061
208	Filippo Inzaghi	-0.031	224	Jerome Arpinon	-0.040	240	Pepe Bordalas	-0.068
209	Bruno Irles	-0.031	225	Chris Wilder	-0.041	241	Christian Streich	-0.115
210	Frederic Antonetti	-0.031	226	Dejan Stankovic	-0.041	242	Sean Dyche	-0.171
211	Alexander Blessin	-0.032	227	David Moyes	-0.041			

Appendix A.2: Regression Table for Interpolation of Wage Data

	Dependent Variable
	WeeklyWageEUR
Age	3,451.084*** (123.279)
Role	Yes
Team	Yes
Constant	-101,073.100*** (12,702.420)
Observations	13,810
R^2	0.479
Adjusted R^2	0.474
Resid. Std. Error	56,881.610 (df=13657)
F Statistic	82.747*** (df=152; 13657)
	*p<0.1; **p<0.05; ***p<0.01

Appendix A.3: Top and Bottom 10 Manager and Position Groups

	Manager	Position	Rating		Manager	Position	Rating
1	<i>Gian Piero Gasperini</i>	<i>Dribbler</i>	0.779	1677	<i>Erik ten Hag</i>	<i>Creator</i>	-0.039
2	<i>Pepe Bordalas</i>	<i>Deep Mid.</i>	0.577	1678	<i>Roberto Donadoni</i>	<i>Midfielder</i>	-0.039
3	<i>Gian Piero Gasperini</i>	<i>Creator</i>	0.516	1679	<i>Walter Zenga</i>	<i>Deep Mid.</i>	-0.040
4	<i>Marcelo Bielsa</i>	<i>Midfielder</i>	0.516	1680	<i>Pepe Bordalas</i>	<i>Midfielder</i>	-0.040
5	<i>Gian Piero Gasperini</i>	<i>Finisher</i>	0.508	1681	<i>Dejan Stankovic</i>	<i>Dribbler</i>	-0.042
6	<i>Jurgen Klopp</i>	<i>Finisher</i>	0.482	1682	<i>Michael Kollner</i>	<i>Finisher</i>	-0.046
7	<i>Simone Inzaghi</i>	<i>Creator</i>	0.476	1683	<i>Christian Streich</i>	<i>Wide Def.</i>	-0.048
8	<i>Jurgen Klopp</i>	<i>Dribbler</i>	0.451	1684	<i>Aurelio Andreazzoli</i>	<i>Finisher</i>	-0.052
9	<i>Pep Guardiola</i>	<i>Finisher</i>	0.423	1685	<i>Sean Dyche</i>	<i>Dribbler</i>	-0.089
10	<i>Pep Guardiola</i>	<i>Cent. Def.</i>	0.353	1686	<i>Sean Dyche</i>	<i>Creator</i>	-0.119

Appendix B.1: Chart of Sean Dyche's Position-wise Performance

